

## ARTICLE

# Using large language models to analyze political texts through natural language understanding

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## Abstract

Large language models (LLMs) offer scalable alternatives to human experts when analyzing political texts for *meaning*, using natural language understanding (NLU). Qualitative NLU methods relying on human experts are severely limited by cost and scalability. Statistical text-as-data methods are scalable but rely on strong and often unrealistic assumptions. We propose a systematic, scalable, and replicable method that can extend existing qualitative and quantitative approaches by using LLMs to interpret texts meaningfully rather than as mere data. Our ensemble means of LLM-generated estimates of party positions on six key issue dimensions correlate highly with equivalent mean ratings by country specialists. When applied to coalition policy declarations, LLM estimates align more closely with standard models of government formation than hand-coded estimates. We conclude with a discussion of the profound implications of modern LLMs for political text analysis.

Reliably extracting information from text is fundamental to social science. This often means identifying authors' stated preferences on key issues such as environmental protection or immigration policy. Written documents and speech transcripts are rich sources of such information, and scholars often systematically describe and compare such preferences using latent issue dimensions. The ideal way to extract preference information from a text involves reading it for meaning and interpreting its issue positions. This fundamentally qualitative approach has been practiced informally by generations of country specialists, using deep familiarity with both the language and the political context of the relevant setting.

Qualitative text labeling is a longstanding research tradition that has produced widely used datasets

in political science, including the Manifesto Project (hereafter MP; Budge et al., 1987, 2001; Klingemann et al., 2006; Volkens et al., 2013) and the Comparative Policy Agendas Project (CAP; Baumgartner et al., 2006, 2019). Both projects spent decades, and millions of dollars, labeling the policy content of political documents—over 3 million sentences and 5000 manifestos for the MP alone—using hundreds of trained expert annotators. Despite widespread use in the profession of datasets produced by the MP and CAP, we have no sense of their reproducibility. These datasets are too expensive to replicate. While alternative methods use non-experts to crowd-source sentence labeling (Benoit et al., 2016; Snow et al., 2008), these also scale poorly, typically requiring dozens of crowd coders to process single documents, even when the

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The Cornell Center for Social Sciences verified that the data and replication code submitted to the *AJPS* Dataverse replicate the numerical results reported in the main text of this article.

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aim is to identify a small target of relevant policy statements. More generally, methods based on human input do not scale.

Seeking ways to extract information from political text at scale, scholars devised and implemented methods that are fully or partially *quantitative*. Automated “text-as-data” (TaDa) approaches had considerable success using quantitative methods to analyze words in a text, despite minimal to zero knowledge of the language or political context involved and remaining completely blind to the text’s meaning (see Benoit, 2020; Gentzkow et al., 2019; Grimmer & Stewart, 2013). In their crudest forms, TaDa methods use statistical techniques to analyze counts of tokens, while more sophisticated approaches may also rely on vector representations of text units (“word embeddings,” e.g., Mikolov et al., 2013).

The key shortcoming of quantitative methods that ignore a text’s meaning is that they struggle to capture its nuances, including the intensity of positions it expresses. Consider two sentences about labor market policy. “Capital is dead labour, which, vampire-like, lives only by sucking living labour, and lives the more, the more labour it sucks.” “We will establish an independent review to consult on how to set a genuine Living Wage across all sectors.” Both set out pro-worker policies. The first comes from *Das Kapital*<sup>1</sup> and uses only a few words to convey a trenchant far-left position in unambiguous terms. The second comes from the UK Liberal Democrat Manifesto of 2019 and sets out a much weaker left-wing position. Any human reader can immediately see the stark difference between these statements. Traditional TaDa methods struggle—if not fail outright—to do this. While it has long been recognized that such methods depend on “wrong” models of language (Grimmer & Stewart, 2013, p. 3), they have been widely accepted, absent a better alternative.

Existing approaches to text analysis therefore face a fundamental trade-off. Quantitative TaDa methods scale to large corpora but treat words as features of the data, often relying on statistical assumptions that abstract away their meaning. In contrast, qualitative human text labeling captures the meaning and intent behind language but lacks scalability. Here, we describe a method of text analysis based on an ensemble of large language models (LLMs). Unlike TaDa models, LLMs are powered by massive pre-training, using effectively the entire Internet, of huge models with billions or even trillions of parameters. This gives them extraordinary ability to summarize and draw inferences from large texts, combining the scalability

and reproducibility of quantitative TaDa approaches with the interpretative nuance of traditional qualitative methods.

Given this, our method first asks LLMs to summarize what a text has to say about particular issues. We then ask them to answer questions about the author’s issue positions. This is a canonical application of “Natural Language Understanding” (NLU), a sub-field of natural language processing focused on tasks such as text summarization and question answering—precisely what we are using LLMs to do here. Note that “understanding” in NLU is not understanding in the sense that machines use anything approaching human cognition. NLU characterizes a set of machine tasks that can *replicate outputs* of human interpretations of text (Samant et al., 2022, review the NLU literature).

Using LLMs in this way offers the scalability of quantitative methods applied to interpretative text analysis—the output of which captures some of the depth and richness of traditional qualitative analysis. This has the potential to revolutionize text analysis in political science, offering cheap, agile, but effective ways to generate results that are functionally equivalent to traditional qualitative text analysis, while doing this at potentially massive scale in practically any language. This has led to their increasing adoption in social sciences where qualitative text analysis has traditionally been the dominant paradigm (see Karjus, 2025; Linegar et al., 2023).

In any scientific endeavor, what is crucial is *not any particular dataset* but a *reliable and replicable method for generating data* that addresses the precise problem at hand. In what follows, we develop protocols for using LLMs in this way and show in our findings that, with scrupulous run design, LLM-based results are reliable, reproducible, and valid.

## USING LLMs TO ESTIMATE ISSUE POSITIONS IN POLITICAL TEXT

Our goal is to develop protocols for systematically and reliably using LLMs to extract information about political preferences of the authors of political texts. Our workflow is summarized in Table 1 and described in the sections that follow.

### Input texts and validation data

**Text corpus:** Party manifestos and coalition agreements

The results we report in this paper concern issue positions in party manifestos and coalition agreements. The same approach can be used for other

<sup>1</sup> Marx, Karl (1976). *Capital: A critique of political economy* (Vol. 1, Chapter 15) (B. Fowkes, Trans.). Penguin Books. (Original work published 1867). This quote was located in *Das Kapital* by ChatGPT as an example of an extreme anti-capitalist position.

**TABLE 1** AI-powered text analysis workflow.

|                                                                                                                                                                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>1. Input texts and validation data</b>                                                                                                                                                                                                              |  |
| a. Text corpus:                                                                                                                                                                                                                                        |  |
| i. Party manifestos from the Manifesto Project (MP)                                                                                                                                                                                                    |  |
| ii. Coalition agreements from Klüver et al. (2023)                                                                                                                                                                                                     |  |
| b. Positions on six issue dimensions from expert surveys fielded in election years:                                                                                                                                                                    |  |
| i. Laver and Hunt (1992); Benoit and Laver (2006)                                                                                                                                                                                                      |  |
| ii. Chapel Hill Expert Survey                                                                                                                                                                                                                          |  |
| c. Issue positions on same six dimensions derived from scaling hand-coded labels:                                                                                                                                                                      |  |
| i. MP data for party manifestos                                                                                                                                                                                                                        |  |
| ii. Klüver et al. (2023) data for coalition agreements                                                                                                                                                                                                 |  |
| <b>2. Large language model (LLM) authorization</b>                                                                                                                                                                                                     |  |
| Establish high-tier API access to three commercial LLMs: OpenAI's GPT-4o, Anthropic's Claude 3.5 Sonnet, Google's Gemini 1.5 Pro                                                                                                                       |  |
| <b>3. Summarize texts</b>                                                                                                                                                                                                                              |  |
| a. Present original language texts to LLMs for processing                                                                                                                                                                                              |  |
| b. Prompt LLMs to generate 200–300-word English language summaries of each text's content relating to each of the six issue dimensions.                                                                                                                |  |
| <b>4. Scale text summaries</b>                                                                                                                                                                                                                         |  |
| Prompt LLMs to locate positions of each summary on the specified 7-point scales                                                                                                                                                                        |  |
| a. for “zero-shot” learning (no examples)                                                                                                                                                                                                              |  |
| b. for “few-shot” learning (adding three sample summaries and associated benchmark positions for each scale)                                                                                                                                           |  |
| <b>5. Validation</b>                                                                                                                                                                                                                                   |  |
| a. <i>Concurrent validity</i> : Correlate ensemble means of LLM estimates of party positions with ensemble means of expert judgments from 1b                                                                                                           |  |
| b. <i>Reliability</i> : As a benchmark for reliability, compare LLM-expert correlations with expert–expert correlations.                                                                                                                               |  |
| c. <i>Predictive validity</i> : Compare ensemble means of LLM estimates of coalitions' issue positions with analogous hand-coded estimates from 1c, to see which better accords with a prediction common to almost all models of government formation. |  |
| <b>6. Replication and robustness checks</b>                                                                                                                                                                                                            |  |
| Repeat Steps 2–5a to see if results are stable.                                                                                                                                                                                                        |  |
| Replicate results using three open-weight LLMs: DeepSeek-V3 (671B), LLaMA-3.3 (70B), and Gemma-3-27B-it.                                                                                                                                               |  |

Notes: This lists each of the steps in the workflow generating the results reported in this paper.

political texts—for example, legislative speeches or social media posts. Substantively, we focus validation of our approach on party manifestos because estimating parties' issue positions is a core research program in political science (Laver, 2003), while manifestos are parties' published commitments to these positions. Methodologically, we focus on parties' issue positions because there are authoritative and widely used datasets, in particular the MP and expert survey series we describe below, dealing with precisely the quantities we wish to estimate. This allows us to validate results in a high-information environment, before moving on to analyze the issue positions of coalition cabinets, on which much less research has been conducted.

Our text corpus for party manifestos is the subset of manifestos in the MP collection (Merz et al., 2016) issued in years for which expert surveys were also fielded. This comprises 235 party manifestos written in 21 different languages (detailed in Supporting Information Appendix A1, p. 3). We downloaded these

from the MP collection, and cleaned and converted them into plain text format for input into LLMs.<sup>2</sup> Our corpus of coalition policy agreements is the set of 23 coalition agreements, from a collection assembled by Klüver et al., for multi-party cabinets forming in election years covered by our manifesto corpus (Klüver et al., 2023).

Validation data: Expert surveys and MP issue scales

Whenever we evaluate a new measurement instrument, a standard way to assess the validity of its results is to compare these with the judgments of experts who have deep local knowledge of the matter under investigation. As a form of “expert crowd sourcing,” published expert surveys are widely accepted as

<sup>2</sup> Since we began this work, most commercial LLMs can read most pdf documents.

**TABLE 2** Expert survey dimensions constant over time.**Economic:** Position on improving public services versus reducing taxes

0 = Strongly favors improving public services

10 = Strongly favors reducing taxes

**Social:** Position on social lifestyle

0 = Strongly supports liberal policies

10 = Strongly opposes liberal policies

**Immigration:** Position on immigration policy.

0 = Strongly opposes tough policy

10 = Strongly favors tough policy

**EU:** Overall orientation of the party leadership toward European integration

1 = Strongly opposed

7 = Strongly in favor

**Environment:** Position toward the environment

0 = Environmental protection even at the cost of economic growth

10 = Economic growth even at the cost of environmental protection

**Decentralization:** Position on political decentralization to regions/localities

0 = Strongly favors political decentralization

10 = Strongly opposes political decentralization

*Notes:* This shows the anchor interpretations and scores for the six expert survey dimensions used for external validation.

reliable and valid estimates of party policy positions. We therefore assess the *convergent* validity of our LLM-generated data by comparing these with qualitative judgments given by ensembles of experts collected in a widely used expert survey series, which we use as our benchmark. This begins with a survey by Laver and Hunt in 1989, extended by Benoit and Laver in the early 2000s (Benoit & Laver, 2006; Laver & Hunt, 1992). The Chapel Hill Expert Survey (CHES) series used the same dimensions and vignettes as these in four expert surveys fielded between 2006 and 2019 (Jolly et al., 2022).<sup>3</sup> We selected six issue dimensions consistently deployed over this series, each concerning a different aspect of policy, for estimation using the LLMs. Table 2 lists these, together with the end-of-scale vignettes offered to experts.

In answering these surveys, expert survey respondents are instructed to rely on their overall knowledge of the parties, of which manifesto content is only a potential component. We also compare our results, therefore, with positions on logit scales<sup>4</sup> referring to the same six issues, derived from published estimates generated by the MP, based on quantitative analyses of the qualitative human labeling of manifesto text units.<sup>5</sup>

To evaluate *predictive* validity, we take as our benchmark a prediction shared by many spatial models of government formation, regardless of the bargaining model on which they are based. This is that the agreed coalition issue position should fall within the range of member parties' issue positions. If it were outside this range, then a Pareto improvement would be possible by moving it into this range.

## Choosing LLMs and software environment

LLMs fall into three broad categories: proprietary, open-weight, and open-source. Proprietary models—such as GPT-4o, Claude, and Gemini—are closed systems: Their training data, model code, and weights are not publicly released. While often the most powerful and user-friendly, they evolve in opaque ways and pose challenges for replicability, transparency, and data control. Open-weight models, including LLaMA 3, DeepSeek-V3, and Gemma-3, release their model weights under restrictive licenses. They can be downloaded and run locally or deployed on private cloud infrastructure. This offers greater transparency and control—especially for research requiring data confidentiality or, more crucially for the social sciences, long-term reproducibility (Weber & Reichardt, 2023). Unlike proprietary models, open-weight models can be extended by fine-tuning or continued pretraining on domain-specific corpora, enabling users to adapt them more directly to their research needs. However,

<sup>3</sup> As of April 2025, the trend file for 1999–2010 CHES data had 1,328 Google Scholar citations (Bakker et al., 2015).

<sup>4</sup> As recommended by Lowe et al. 2011, see Supporting Information Appendix A2 (p. 4) for details.

<sup>5</sup> As of April 2025, the MP website listed 785 peer-reviewed publications using these data, over and above 63 publications by the MP core team.

they do not typically disclose their training data or code and thus are not fully open-source. Fully open-source models (such as Pythia, Mistral, or Falcon) go further, releasing not only model weights but also training data and training code under permissive licences. While this is useful for understanding and manipulating the details of how LLMs work, open-source models are much less powerful, with much less language coverage, than the best proprietary and open-weight models, so we do not consider them here.

Privileging powerful and accessible text analysis at scale, and designed to analyze public domain documents, our baseline method uses three proprietary LLMs: GPT-4o (from OpenAI), Gemini 1.5 Pro (from Google), and Claude 3.5 Sonnet (from Anthropic). Free browser-based versions of these are impractical for the text analysis workflow outlined in Table 1.<sup>6</sup> Subscription versions give access to the model's Application Programmer Interface (API), allowing much more powerful analysis to be integrated into the user's software.<sup>7</sup> In light of the increasing performance of the best open-weight models, our replication analysis also used three of these: DeepSeek-V3-0324; LLaMA-3.3-70B-Instruct (from Meta); Gemma-3-27b-it (from Google).<sup>8</sup>

LLMs typically have a *temperature* setting controlling the randomness—or “creativity”—of generated text: higher values increase variability by allowing the model to sample less probable tokens when generating responses. To maximize replicability, we set the temperature to zero for all models, which encourages more deterministic outputs but does not fully eliminate variability. We also set the *top-p* (nucleus sampling) parameter to 1.0 to prevent additional sampling constraints, and where available, we fixed the *random seed*. Despite these measures, we observed small residual variation in some models' outputs across runs—likely due to stochastic processes within their generation pipelines or undocumented aspects of inference architecture.

The systematic analysis of large text corpora using multiple prompts and a variety of LLMs requires a robust software environment. This is provided by Langchain (Chase, 2022), an open-source package that is a central hub for the development of LLM applications. This allows users to run the same analysis on the latest versions of a wide range of LLMs—proprietary as well as open-weights and open-source. Langchain

also requires specification and recording of the precise issue number of each LLM deployed in each analysis—crucial replication information often lost or omitted in reports of more casual LLM-based analyses.

We are not the first to use GPT-class LLMs to estimate policy positions from party manifestos. Applying LLMs to tasks from Benoit et al. (2016), both Le Mens and Gallego and Ornstein et al. used LLMs to replicate crowd-sourced and expert classifications of policy content at the sentence level (Le Mens & Gallego, 2025; Ornstein et al., 2025). Bol and Bono simply asked ChatGPT to answer the CHES expert surveys questions but did ground these in manifestos and applied them only to eight French parties for one election (Bol & Bono, 2025). Our approach differs significantly. Rather than analyzing individual sentences and aggregating results into an index, we treat texts *holistically*, asking LLMs to “read” and summarize their policy content in its entirety before scoring the summarized content on issue scales. While the LLMs do not replicate human cognition in any mechanistic sense, their capacity to interpret a document as a coherent whole enables a *functional equivalence* to how a human coder might read and assess a text's policy positions on particular issues. Instead of relying on human qualitative coding or using LLMs as a machine learning tool for sentence-level classification, therefore, we exploit their NLU capacities for abstractive summarization, interpretation, and question answering (Zhang et al., 2024).

## Summarizing texts using LLMs

The main proprietary LLMs can analyze very long texts. However, it is neither efficient nor effective to feed an LLM a large text (some manifestos exceed 150 pages) and ask it to rate positions on an issue discussed in only a sentence or two. This is the “needle in the haystack” problem (Chang et al., 2023; Hosseini et al., 2024; Wang et al., 2024). We modularized the huge transition from a large text to a single-integer summary of an issue position by leveraging a capability at which LLMs are highly effective: generating concise summaries of large documents (Hoyle et al., 2023; Wu et al., 2024). We therefore introduced an intermediate stage between the full text and the single-integer summary, asking the LLMs to generate a short English-language summary of what the original language manifesto says about the issue in question. We engaged in extensive reading of these summaries, which were typically lucid and on-topic, to check their substantive validity. Value added by the summary stage became clear when we prototyped a method bypassing this, which resulted in substantially worse results.<sup>9</sup>

<sup>6</sup> Browser versions are also increasingly “agentic,” unlike models accessed via the API, making web calls on the fly rather than relying solely on their training data. This makes their output less replicable.

<sup>7</sup> LLM APIs typically have tiers; higher tiers have increasing costs but much higher volumes and rates of inputs and outputs. It was essential in our work to spend (quite modest) funds to move up several tiers for each LLM.

<sup>8</sup> Given the considerable computing power required, we found it most convenient to run these on virtual machines hosted by the Nebius platform (<https://studio.nebius.com/>), but with sufficiently powerful hardware, all three LLMs could have been deployed locally. In all three cases, we were able to download and archive the model files for the open-weight LLMs.

<sup>9</sup> See Supporting Information Appendix B, especially Table B5 (p. 15).

Our prompt strategy for using the LLMs to summarize long texts was extensively explored during prototyping, described in Supporting Information Appendix B, pp. 12–17. We experimented with multi-issue summaries as well as longer or shorter summaries and settled on prompting for six short (300–400 words) single-issue summaries per document.<sup>10</sup>

We see generating intermediate text summaries as an integral part of our method, even as LLMs continue to advance in capabilities. Improvements in text summarization will naturally accompany these advances, but passing large volumes of text to an LLM still carries the risk of context or attention drift (Liu et al., 2023). More importantly, even if this issue were completely resolved, using English language summaries provides critical transparency—especially for researchers working with texts written in languages they cannot understand. These summaries offer a traceable representation of the input content, enabling human review and validation in a way that would be lost if the process moved directly from a 150-page original-language document to a single numerical score. Crucially, this transparency also permits us to inspect and validate how the LLMs derived their positions, something we illustrate below through a closer examination of the decentralization and environmental policy domains.

The summarizer performs two tasks simultaneously: It summarizes the original-language manifesto and returns that summary in English. During prototyping, we explored whether performance could be improved by modularizing these tasks—first translating the manifesto, then summarizing the translation. However, we found that correlations with expert survey scores tended to be lower when a translation stage was inserted before summarization,<sup>11</sup> so we excluded this step from our final design.

## Scoring issue positions in text summaries

Concerned with the possibility that small changes in the wording of scoring prompts might have big effects on LLM outputs, following Battle and Gollapudi (2024), we tested a matrix of nine different wordings for each LLM scoring prompt.<sup>12</sup> We concluded that small changes in prompt wording of the type we investigated are not critical in this application, settling on the prompt wording most closely corresponding to prompts given by CHES to human experts: “You are an expert social scientist with a PhD in political science. Think carefully about your answer.” Again, responding to the received wisdom that LLMs perform better, the more help they are given, we settled on a standard 7-point scale for recording issue preferences, supply-

ing detailed substantive interpretations of each of the seven points for each of the six issue scales. These are set out in Supporting Information Appendix A4, pp. 6–11.<sup>13</sup>

## Few-shot learning

Asking LLMs to score texts without giving examples of what a “good” score looks like is known as “zero-shot” learning. “Few-shot” learning, by contrast, uses longer LLM prompts that include sample text summaries along with associated benchmark scores. We implemented few-shot learning, using benchmark scores from expert surveys. To generate training examples, we selected manifesto summaries—edited for succinctness and clarity—from manifestos associated with expert survey scores that span the issue dimensions of interest. For each issue, our few-shot scoring prompts supplement the zero-shot prompts with three examples—from the right, left and center of each dimension—paired with their corresponding expert survey scores.<sup>14</sup>

## Run design and ensemble means

Following design decisions discussed above, the final “3 × 3” run design for our baseline text analysis was as follows. For each of the 235 manifestos in Supporting Information Appendix A1 and each of the six dimensions in Table 2, we first used each of Claude, GPT, and Gemini to generate 300- to 400-word summaries (in English) of the issue positions set out in the full native language text. This generated  $235 \times 6 \times 3 = 4,230$  issue position summaries. We then used each of Claude, GPT, and Gemini to score each of the three summaries per manifesto-dimension on a 1–7 scale, returning “NA” if there was insufficient information. This generated nine LLM scores (including NAs) for each issue dimension for each manifesto—12,690 LLM scores in all. We used this design for two different runs, the first with zero-shot learning and the second with few-shot learning, as specified above. The total output was an ensemble of 18 LLM scores for each manifesto-dimension—25,380 LLM scores in all.

We use the mean of the 18 estimates from zero- and few-shot 3 × 3 runs to aggregate results for each issue-manifesto generated by our ensemble of LLMs (Abburi et al., 2023; Sujatha & Nimala, 2024). This offers a robust and simple way to compare results of our manifesto analysis, using multiple LLM

<sup>10</sup> For prompt wordings, see Supporting Information Appendix A3 (p. 5).

<sup>11</sup> Detailed results in Supporting Information Appendix Table B7 (p. 17).

<sup>12</sup> See Supporting Information Appendix Table B1 (p. 12).

<sup>13</sup> We conducted an initial prototyping run using the CHES 11-point scales, which have two anchors, one at each end. This produced systematically worse results than the 7-point, 7-anchor scales we developed, likely because of the precise meaning given to each scale position.

<sup>14</sup> Since the LLMs are asked to return integer scores, we rounded the relevant rescaled CHES means.

summarizers and scorers, with the ensemble means in benchmark expert survey data (Yang et al., 2025).

## SUBSTANTIVE VALIDITY OF LLM ESTIMATES OF PARTIES' ISSUE POSITIONS

### Benchmarks for substantive validity

Our quantity of interest is the “true” position of party  $p$  on issue dimension  $d$ . Given a set of parties and issues, we want to estimate a matrix of parties' true issue positions,  $T_{pd}$ . These positions may mean many things. For our purposes, they are positions motivating party leaders' elite actions in government, opposition, or government formation. These true positions exist only in the minds of politicians whose observable behaviors may well be strategic.  $T_{pd}$  is *fundamentally unobservable*. This means that there is no observable “ground truth” target with which we can compare our estimates—a problem common to many methods of data collection in the social sciences, including all survey research.

We use our ensemble of LLMs to generate  $L_{pd}$ , a new observable *estimate* of unobservable  $T_{pd}$ . To validate  $L_{pd}$ , we compare it with two published data sources, the expert survey series and the MP. Expert surveys ask a collection of experts to score party positions on issue dimensions with substantively defined endpoints.<sup>15</sup> For a given setting and time point, an expert survey generates an observed matrix,  $E_{pd}$ , of party positions on dimensions of interest. Each cell in the matrix is an ensemble mean of expert scores. There is substantial variation around these means. This arises both from disagreements among experts on the “true” position and the varying frames of reference they might bring to bear on their task.<sup>16</sup>  $E_{pd}$  is an observable estimate of  $T_{pd}$ .

Our conjecture is that observed expert survey and LLM estimates,  $E_{pd}$  and  $L_{pd}$ , are both systematically related to our unobservable target,  $T_{pd}$ . We therefore expect them to be systematically related to each other. Using  $E_{pd}$  as our benchmark, therefore, we assess the content validity of  $L_{pd}$  using correlations between  $L_{pd}$  and  $E_{pd}$ . It is also possible that, notwithstanding high correlations,  $L_{pd}$  is *biased* relative to  $E_{pd}$ —it is impossible to know whether it is biased relative to  $T_{pd}$  since the latter is unobservable. We investigate this by comparing *distributions* of scores generated by  $L_{pd}$  and our  $E_{pd}$  benchmark.

<sup>15</sup> The median number of experts per party was 16 in the Benoit–Laver survey and 12 in the CHES series.

<sup>16</sup> The median standard deviation of expert estimates of party positions on the 1–20 scales used in the Benoit–Laver survey was 3.3, and the median standard error .75. Given this level of variance in the expert estimates, a test of whether an LLM could have been an expert in an expert survey would present a very soft target.

**TABLE 3** Upper bounds of human reproducibility: Pearson correlations between split-sample expert surveys.

| Issue dimension  | Mean | 95% CI     |
|------------------|------|------------|
| Economic         | .88  | [.85, .90] |
| Social           | .91  | [.89, .93] |
| Immigration      | .88  | [.86, .91] |
| EU               | .95  | [.93, .96] |
| Environment      | .84  | [.82, .87] |
| Decentralization | .78  | [.74, .82] |

Notes: 95% CI denotes the 95% confidence interval.

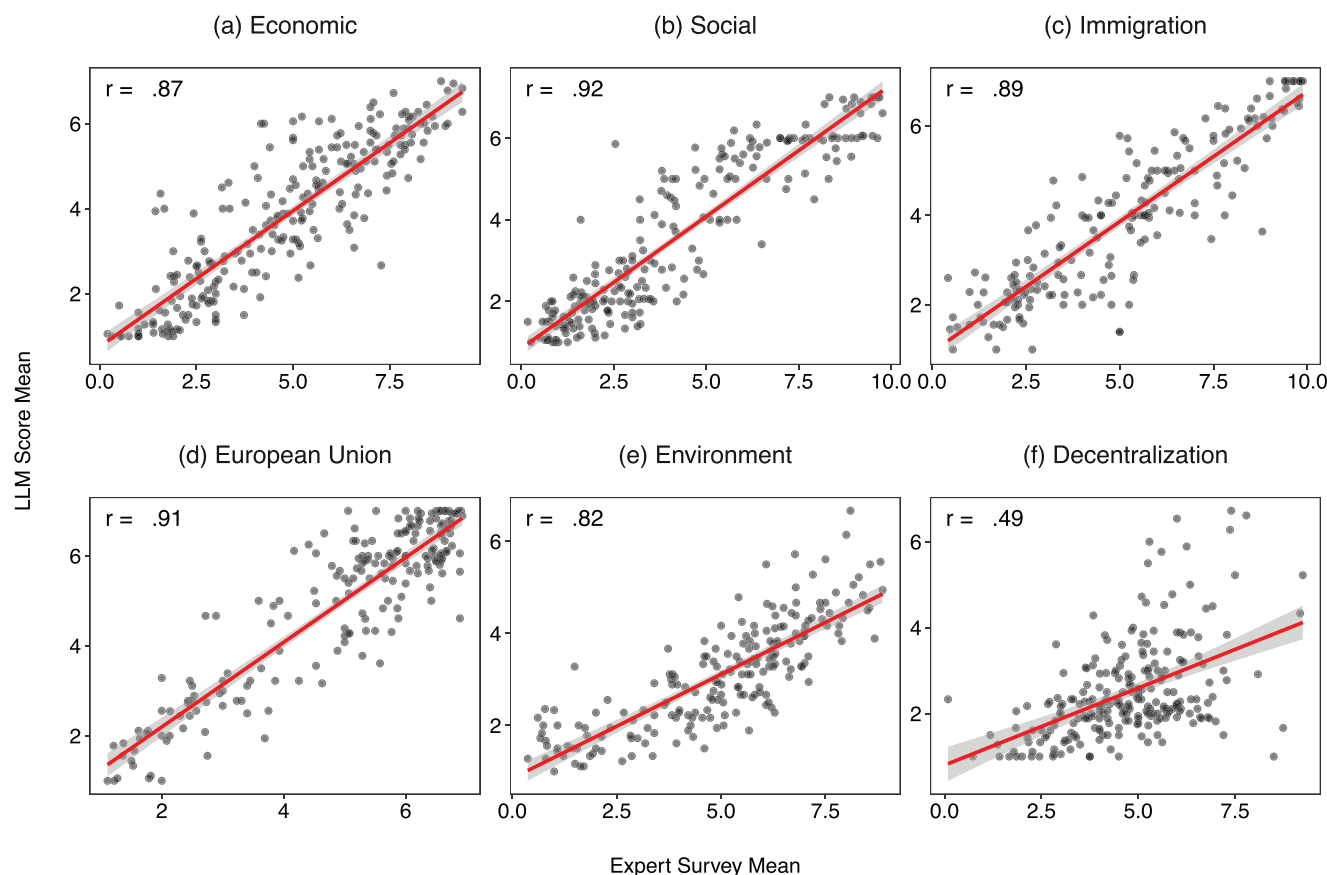
$L_{pd}$  is generated by using LLMs to analyze party manifestos, to which experts generating the  $E_{pd}$  benchmark are never referred. We therefore also compare LLM estimates with data from the MP, which, unlike expert survey data, explicitly derive from manifesto content. We computed MP estimates of party positions on each issue dimension using logit scales recommended by Lowe et al. (2011).<sup>17</sup> These generate a manifesto-based observed data matrix,  $M_{pd}$ . The value of comparing our LLM estimates with  $M_{pd}$  as well as our  $E_{pd}$  benchmark, as we do below, is that there may be issues for which party manifestos *per se* are “biased” relative to expert judgments—biases in parties' *published* issue positions that have nothing to do with measurement instruments.

### Upper bound on correlations between LLM and expert ensemble means

Our expert survey benchmark is an ensemble mean of quantities with substantial variation, arising both from sample variability among expert respondents and fundamental uncertainty about the meaning of policy scales and positions (see Benoit & Laver, 2006). This variation sets an upper bound on correlations between LLM and expert survey estimates that can reasonably be considered a benchmark for success. To establish this, we simulated correlations between two expert surveys estimating the same issue positions, using the Benoit–Laver replication dataset.<sup>18</sup> For each country, we randomly split the expert pool into two sub-samples. For each sub-sample, we calculated mean expert scores for each quantity of interest. We then correlated the resulting pairs of expert means, simulating the correlation between two expert surveys. Repeating this 1,000 times generates *distributions* of expert–expert correlations, with the means and ranges reported in Table 3. This shows that, given typical variation in expert responses, mean correlations between estimates of the same issue positions by different

<sup>17</sup> For details, see Supporting Information Appendix A2 (p. 4).

<sup>18</sup> Chosen because this involves the largest number of respondents in the expert survey series.



**FIGURE 1** Plot of large language model (LLM) ensemble of 18 scores versus expert means, by issue. *Notes:* Each point is a party manifesto, plotting the mean expert score (horizontal axis), against the LLM ensemble mean (vertical axis) for the same document.

expert surveys are typically around .90—somewhat lower for decentralization and environmental policy. We cannot reasonably expect LLM estimates to correlate with the expert benchmark at a level higher than this upper bound.

### Results: Observed correlations between LLM estimates and expert benchmark

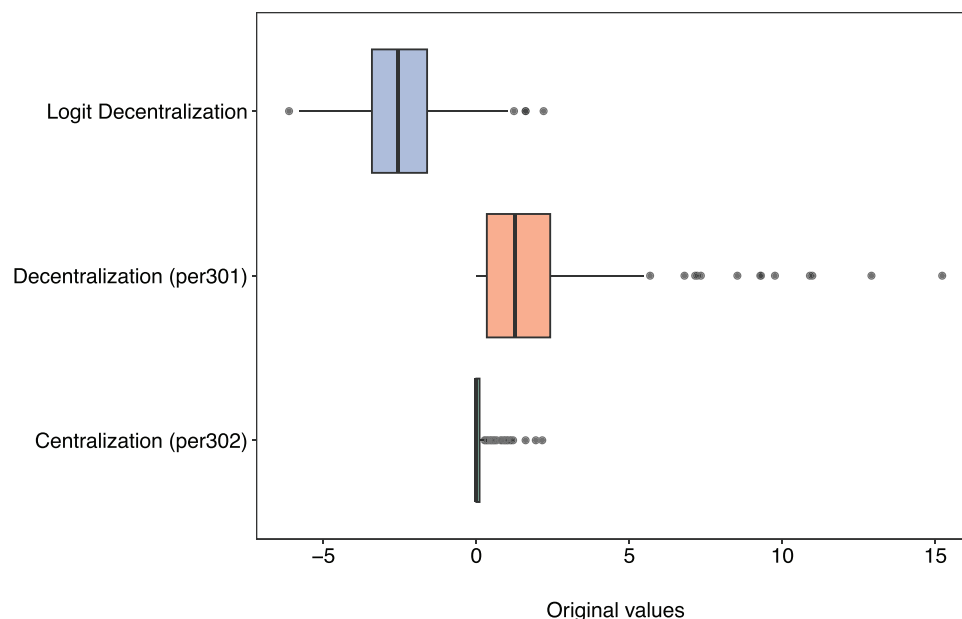
We assess the content validity of LLM estimates of party policy positions in terms of their correlations with benchmark expert survey estimates of positions of the same parties on the same issues. Figure 1 summarizes these results, plotting for each issue the relationship between the ensemble mean of LLM scores and the mean expert survey score. Pearson correlations between the LLM ensemble scores and benchmark expert survey means are high for most issues, often ranging from .87 to .92. Except for decentralization—to which we return in detail below—these results are far better than we hoped for at the beginning of this project. They are at or close to the upper bound, identified in Table 3, of what can reasonably be expected. This is compelling evidence of

the convergent validity of our LLM-based approach for estimating party policy positions.

Detailed results, reported in Supporting Information Appendix Table C1 (p. 18), show these correlations to be robust between different summarizing and scaling LLMs. Supporting Information Appendix Tables C1 and C2 (p. 19) show that correlations between few-shot LLM and expert scores are essentially the same as for zero-shot scores. For many issues, zero-shot correlations are already close enough to the upper bound that there is very little headroom for improvement with few-shot results.

### Decentralization: Do experts see issue positions not found in party manifestos?

Figure 1 shows lower correlations between LLM and expert scores for the decentralization dimension. However, we found that the decentralization scores from different LLMs are highly correlated with each other (Supporting Information Appendix Table C3, p. 20). This suggests that different LLMs consistently identify the same patterns, raising the question of whether experts are basing their judgments



**FIGURE 2** Distributions of Manifesto Project (MP)-estimated manifesto percentages on (de)centralization for manifestos in the LLM corpus. *Notes:* This boxplot of percentages of mentions measured by the MP shows that almost no manifesto has quasi-sentences advocating the centralization of decision-making.

on information not present in the published party manifesto.

As we show in Figure 3, expert scores on decentralization span the full range of the policy scale. An informal reading of the manifesto summaries, however, reveals that nearly all party manifestos publicly advocate some form of decentralization, with almost none opposing it. A similar, though less pronounced, pattern occurs with environmental policy: Experts score party positions across the full scale, yet most manifestos publicly promote relatively pro-environmental policies; very few take an opposing stance.

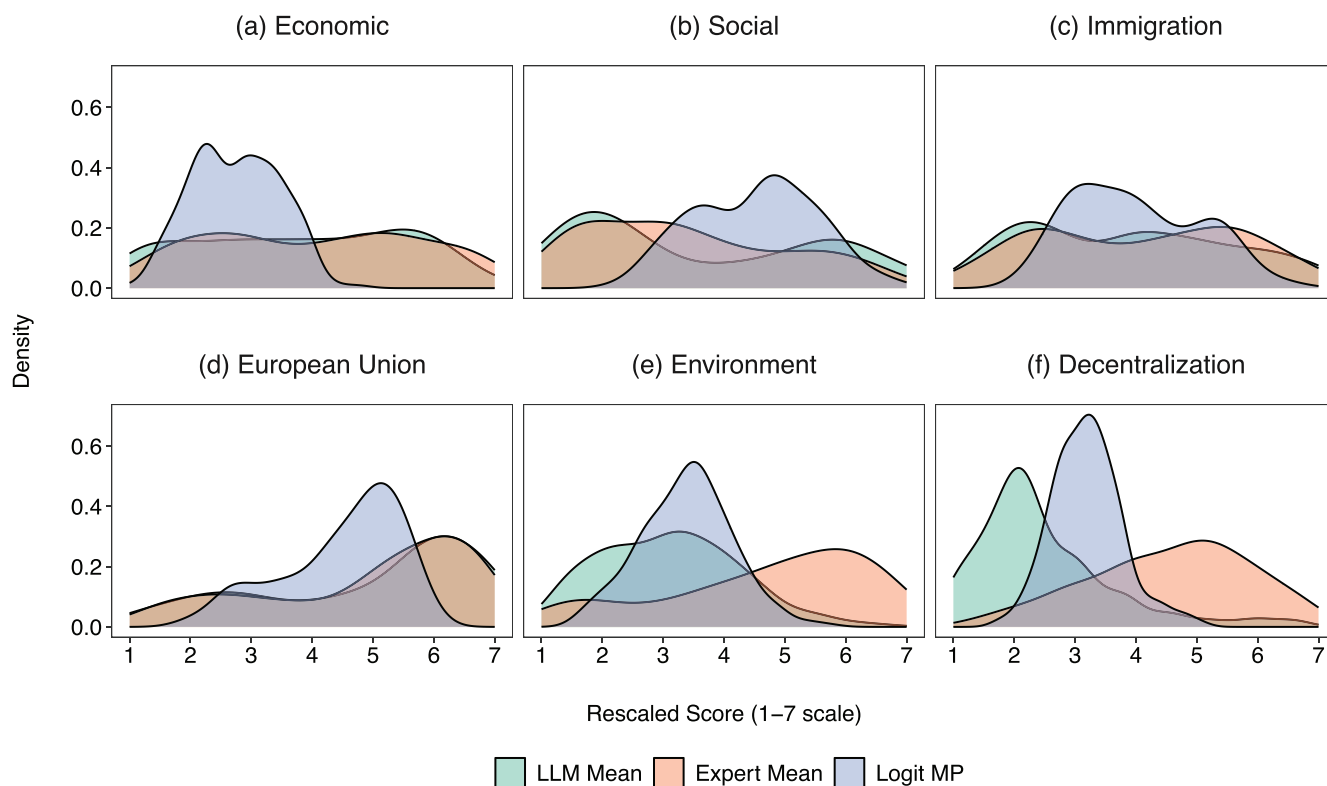
MP data add systematic evidence to the casual observation that the manifestos we analyzed *contain almost no anti-decentralization content*. The MP coding scheme has variables for pro- and anti-decentralization.<sup>19</sup> Figure 2 shows box plots of percentages of MP-labeled quasi-sentences favoring and opposing decentralization for the set of manifestos we analyzed with the LLMs, confirming the observation that these have very few anti-decentralization labels. LLM scores have lower correlations with expert judgments on this issue because, while experts judge parties to have a wide range of decentralization policies, there is almost no observable variation in manifesto positions on decentralization. This lack of variation results, as Tables 5 and 6 show, in lower inter-LLM correlations for decentralization than for the other five issue dimensions. In effect, not only are experts likely seeing features of decentralization policy that are not

expressed in party manifestos, but also the lack of variation in manifesto discussions of decentralization makes this a more difficult information retrieval task for the LLMs.

More generally, we can check the extent to which experts and manifestos say different things about party policy by comparing LLM estimates with MP data and expert scores for the same parties' positions on the same policy dimensions.<sup>20</sup> Figure 3 plots the distributions of these three sets of scores, by issue. The disjunctions between LLM (in green) and expert scores (in red) for decentralization and environmental policy suggest strongly that they are not measuring the same thing. The distribution of MP scores (in blue) for these same dimensions suggests that LLMs are capturing what is in the manifestos on decentralization and the environment. Experts must be reading more into party policy on decentralization and the environment than is published in the manifesto. One way of looking at this is to see manifesto content as structurally “biased, systematically downplaying pro-centralization (and anti-environment) positions for strategic reasons, with expert respondents in effect correcting for this. Which measures are biased with respect to unobservable “true” party policy positions is not a question of bias in LLMs but rather a question of whether parties bias their manifesto commitments on some types of issue dimension, a matter on which Figure 3 offers systematic evidence.

<sup>19</sup> MP codes, PER 301 and 302.

<sup>20</sup> We make direct comparisons between LLM, MP, and expert scores by first rescaling expert and MP scores from the 1 to 7 scale used by the LLMs.



**FIGURE 3** Distributions of LLM 18-score ensemble, expert and MP scores, by issue. *Notes:* Each panel shows, for the relevant issue dimension, a kernel density plot summarizing the distribution of scores estimated by: the LLM ensemble; the expert survey benchmark; a logit scale derived from MP data.

On the remaining four issues—economic, social, immigration, and EU policy—Figure 3 shows close correspondence between distributions of LLM and expert scores. This suggests both are measuring the same thing, reflected in very high correlations between LLM and expert scores for these dimensions. In contrast, the distribution of human-labeled MP estimates does not match LLM or expert scores for these four dimensions. This reflects not only the structural bias from some “valence” issues as noted above but also the limitations of the label-counting approach to aggregating issue positions. In the results shown in Supporting Information Appendix D (p. 21), we note that LLM results correlate much more highly with benchmark expert survey estimates on each of the six policy dimensions than do MP-based estimates.

## ROBUSTNESS AND REPRODUCIBILITY ISSUES

### “Informative missingness” in LLM summarization and scoring

A crucial feature of our LLM scoring prompt is the instruction to return *NA*, or “not applicable,” if the manifesto summary does *not* state a position on the

issue in question. This is crucial as both a matter of principle and to mitigate the well-documented tendency of LLMs to produce “hallucinations” (Bruno et al., 2023; Chrysostomou et al., 2024)—instances where the LLM generates content that appears plausible but is unsupported by, or even contradicts, the source text. By ensuring the model identifies when sufficient information to assign a score is unavailable, we avoid forcing it to generate spurious scores on dimensions lacking evidence in the manifesto. Conversely, when sufficient information is present, we expect the LLM to return a valid and reliable score.

“NAs” returned by LLMs, expert survey respondents, or MP text coders are highly informative. Knowing what is *not* in a text is as important as knowing what is in it. Expert survey respondents often decline to rate a given party on a given dimension when they lack sufficient information. These “informative NAs,” however, are typically not reported in expert survey data comprising ensemble mean scores on party positions. However, they can often be retrieved from replication datasets. For example, 58 experts responded to Benoit and Laver (2006) on issue positions of two UK parties: the well-known Conservative Party and the less well-known *Plaid Cymru* (PC, Welsh Nationalists). For the Conservatives, the number of experts giving judgments on the various issue dimensions deployed

ranged from 54 to 58. For *PC*, this number ranged from 29 (on immigration policy) to 44 (on the economic policy dimension). Does the 50% “expert NA” response on *PC*’s immigration policy mean that these experts judged *PC* to have *no immigration policy* in 2002? Or does it mean that *PC* may have had an immigration policy, but the experts do not have the information or ability to make judgments on this? Both are valid and informative expert responses, with very different substantive meanings.

Similarly, LLMs might return an “NA” issue score for one of two reasons: either the manifesto contains no stated position on a particular issue, or the information does exist, but the LLM could not retrieve it. We gain insight into this by comparing very different rates of NAs returned when we ask LLMs to generate scores directly from large documents without first summarizing these (the most difficult retrieval task), with those when we give LLMs help with retrieval by first asking them to summarize manifesto content on the issue in question. As reported in Supporting Information Appendix B (pp. 15–16), the more difficult retrieval task generates far higher levels of NAs from the same text, suggesting that many of these result from retrieval problems. This supports a conjecture, reinforced by our extensive reading of manifesto summaries, that NAs returned by the much easier task of scoring summaries imply absent or unclear issue positions.

A systematic feature of our results is considerable variation between LLMs—indeed the only systematic difference we found between them—in the frequency of NAs they return for a given issue. One summarizer (typically GPT) may tell us there is policy content on some dimension while another (typically Claude and/or Gemini) tells us there is not. Reading many summaries, we found a tendency for GPT summaries to say something like “there is no explicit mention of X in the manifesto, however the overall ideological orientation of the manifesto suggests that its policy on X is y.” Claude and Gemini summaries were more likely to stop at “there is no explicit mention of X in the manifesto,” which then generates an “NA” at the scoring stage. Table 4 summarizes these differences by computing the proportion of missing scores underlying each of our ensemble means of (up to) 18 zero- and few-shot estimates of the same quantity, by issue dimension and summarizer model.<sup>21</sup> This shows both that GPT-4o summaries generate substantially less missingness and that there is more missingness on some dimensions (e.g., immigration) than others (e.g., the environment).

This highlights a crucial trade-off when we use LLMs to analyze political texts, a trade-off common to most

**TABLE 4** Proportion missing LLM scores contributing to ensemble mean.

| Issue dimension  | Proportion of missing LLM scores |        |                |         |
|------------------|----------------------------------|--------|----------------|---------|
|                  | Claude 3.5                       | GPT-4o | Gemini 1.5 pro | Overall |
| Economic         | .13                              | .02    | .08            | .08     |
| Social           | .24                              | .21    | .38            | .28     |
| Immigration      | .42                              | .24    | .27            | .31     |
| EU               | .16                              | .03    | .05            | .08     |
| Environment      | .12                              | .01    | .04            | .06     |
| Decentralization | .20                              | .01    | .13            | .11     |
| All Issues       | .21                              | .09    | .16            |         |

Notes: Each cell shows, for the respective issue dimension and LLM, the proportion of manifestos for which the LLM returned a “missing” score.

data generation exercises. If we want to be very confident in our estimates, then we may opt to discard estimates for which not all summarizers and scorers agree. The price will be a large amount of missing data. If we want to increase the volume of “non-missing” data, we may tolerate less consensual scores, possibly arising because LLMs disagree on whether there is relevant policy content in the manifesto. This is directly analogous to a decision we might make to use only pristine expert survey scores on which *all surveyed experts both make a judgment and agree* or, in common with most of the profession, to use mean scores based on varying numbers of missing expert responses, knowing that some are more reliable than others.

As we have seen, lower levels of missingness generated by GPT-4o arise because it is more likely to use other manifesto text to infer manifesto content on an issue which is not explicitly mentioned. Whether or not this is a good thing is a decision for the end-user, faced with a trade-off between pristine estimates and missing data, which is common to most measurement exercises. This is an important feature of research design, which we inform here but do not pre-judge. Our best results, however, came from aggregating multiple LLM results into an ensemble mean, just as previous approaches based on experts have found the best estimates to come from aggregating the individual expert scores into a single ensemble mean (see Benoit & Laver, 2006, Appendix A).

## Reliability of LLM policy scores

Our discussion of LLMs’ varying judgments about whether texts contain specific policy content highlights a key point: Just like human experts, LLMs do not always agree. We argue that the workflow outlined in Table 1 produces estimates that are not only valid but also reliable. We examine this claim by analyzing both intra- and inter-LLM agreements.

<sup>21</sup> For a given summary, there was very little difference between LLMs in missingness at the scoring stage.

**TABLE 5** Agreement rates of LLM positional ratings.

| Issue dimension  | Intra-LLM |                | Inter-LLM |                |
|------------------|-----------|----------------|-----------|----------------|
|                  | ICC       | Kripp $\alpha$ | ICC       | Kripp $\alpha$ |
| Economic         | .90       | .90            | .91       | .90            |
| Social           | .95       | .91            | .97       | .91            |
| Immigration      | .93       | .93            | .93       | .92            |
| EU               | .92       | .87            | .92       | .88            |
| Environment      | .80       | .79            | .76       | .74            |
| Decentralization | .66       | .62            | .69       | .57            |
| Overall          | .91       | .90            |           |                |

Notes: ICC is the Inter-class correlation coefficient; “Kripp  $\alpha$ ” is Krippendorff’s alpha for ordinal data.

Leveraging the reproducibility analysis discussed below, Table 5 shows levels of both intra- and inter-LLM agreements for each policy dimension and overall, using standard measures for assessing the reliability of human raters (Burla et al., 2008). We measure how well each LLM agreed with itself in a repeat scoring of the same manifesto, and agreement between different LLMs on the same issue score. Table 5 shows both the interclass correlation coefficient (ICC), which measures agreement rates for interval data, as well as a version of Krippendorff’s alpha designed for ordinal scales (Krippendorff, 2018).

Table 5 shows very high reliabilities, compared to human raters: on both measures typically .90 or above for both intra- and inter-LLM reliability on the first four policy dimensions. The results for the more valence-oriented issues of the environmental and decentralization policy were lower—between .76 and .80 (ICC) for environmental policy and .66 and .69 (ICC) for decentralization policy—though still high, compared to human benchmarks. In experiments with multiple trained coders applying the MP scheme, Mikhaylov et al., 2012, p. 90) found human coders were highly prone to misclassification and error, with reliability scores typically ranging between .3 and .5. Human experts have a “temperature” far from zero; their training data, ability to follow instructions, and internal reasoning processes vary in ways that are wide-ranging and unknowable. LLMs, by contrast, offer remarkably consistent evaluations of textual content, which, while not in perfect agreement, are substantially more so than we expect from human readers doing the same job.

## Addressing potential concerns about “data leakage”

A significant concern when using LLMs to estimate policy positions is *data leakage*—which could arise when a model draws on information it memorized during training rather than interpreting the input text. For instance, an LLM might recall a party’s EU position

from training data instead of inferring it from a manifesto summary, inflating performance and overstating the model’s ability to simulate expert judgments. We took several steps to assess and minimize this risk.

First, neither the data we supplied to the LLMs for positional scaling nor the expert benchmarks used for evaluation were part of the models’ training data. The policy summaries were created specifically for this study and could not have been seen by the models during training. Our prompts explicitly instructed the LLMs to base their answers solely on these summary texts, with no reference whatsoever to benchmark datasets (CHES or others). While LLMs are trained on a wide range of general content, including academic literature on expert surveys, they do not have access to the expert survey datasets themselves.<sup>22</sup> Furthermore, the coalition policy agreement analyses reported in the next section could not be part of any training data, as the issue position estimates we compute as benchmarks for these have never been published. Finally, the 7-point, 7-anchor, scales we asked LLMs to use differ substantially from those used in the benchmark datasets (11-point for CHES, 20-point for Benoit–Laver and Laver–Hunt, in each case with just two anchors), making direct reproduction of benchmark values unlikely, even if somehow the models had encountered them during training.

Second, if LLMs were simply regurgitating memorized expert survey scores, we would expect strong correlations across all dimensions, including decentralization. Instead, decentralization shows substantially lower correlations with expert benchmark across all six LLMs we deployed. This pattern, along with the alignment between LLM estimates and manifesto-based scores from the MP shown in Figure 3, provides strong evidence that the language models are not retrieving benchmark issue positions from their training data but instead are basing their answers on interpreting the input summaries.

Finally, as a robustness check against possible cueing effects from party identification, we created an edited version of each summary in which all party names and abbreviations were removed. Since a summary cannot trigger memorized knowledge about a party’s policy positions if it is not identifiable, this anonymization provides a strong test for leakage. Just under half of the original summaries contained such identifiers. We used these anonymized summaries to generate a new set of issue position scores (using GPT-

<sup>22</sup> The LLMs themselves support this claim. When we asked Gemini about this, it replied: “I would not be ‘drawing on’ or ‘recalling’ specific CHES scores for that party from my training data to answer your questions about the manifesto. My analysis would be grounded in the textual evidence of the manifesto itself. The knowledge about CHES and similar frameworks helps me understand the *type* of questions you’re asking and the *nature* of the policy dimensions, but the content of my answer about the manifesto’s positions would be derived from the manifesto.” See Supporting Information Appendix H (pp. 25–30) for full details of our chats with three LLMs on this matter.

**TABLE 6** Reproduction and replication of LLM-generated results.

| Issue dimension  | Reproducibility                        |                                 |          | Replicability                                  |                                 |          |       |
|------------------|----------------------------------------|---------------------------------|----------|------------------------------------------------|---------------------------------|----------|-------|
|                  | Proprietary LLMs                       |                                 |          | Open-weight LLMs                               |                                 |          |       |
|                  | Second ensemble with original ensemble | Correlation with expert surveys |          | Open-weight ensemble with proprietary ensemble | Correlation with expert surveys |          |       |
|                  |                                        | Replication                     | Original |                                                | LLaMA                           | DeepSeek | Gemma |
| Economic         | .97                                    | .87                             | .87      | .95                                            | .84                             | .84      | .86   |
| Social           | .97                                    | .92                             | .90      | .95                                            | .87                             | .87      | .86   |
| Immigration      | .96                                    | .89                             | .89      | .93                                            | .86                             | .89      | .89   |
| European Union   | .98                                    | .91                             | .91      | .96                                            | .86                             | .86      | .84   |
| Environment      | .92                                    | .82                             | .76      | .92                                            | .68                             | .79      | .86   |
| Decentralization | .69                                    | .49                             | .40      | .78                                            | .40                             | .45      | .45   |

Notes: All values shown are Pearson product-moment correlations. The open-weight models are LLaMA-3-70B, DeepSeek-V3 (671B), Gemma-3-27B-it.

4o and zero-shot learning only) and compared these with the originals. The results were virtually identical, a correlation of .99, indicating that the presence of party names had no meaningful influence on the model's outputs. (Full results are shown in Supporting Information Appendix I, pp. 31–33.)

In conclusion, while data leakage is a legitimate concern when benchmarking LLMs' scoring of issue positions, the lack of access of LLMs to expert datasets, the consistently lower correlations for decentralization, and our robustness checks using anonymized summaries combine to provide strong evidence that our results are not meaningfully affected by leakage.

## Replicating the LLM analyses

Concern has been expressed (among others, by Barrie et al., 2025) about the potential for a “replication crisis” in research based on LLMs—that LLMs, for various reasons, do not provide the machinery for researchers to replicate the data they generate. In political science, the term *replication* is often used to describe what other disciplines refer to as *reproducibility*—that is, the ability to obtain identical results by re-running the original code and data through the same computational pipeline. By contrast, the more general scientific usage of *replicability* refers to obtaining consistent results using different tools or data under similar research conditions (Breuer & Haim, 2024; National Academies of Sciences, et al., 2019, p. 6). We assess reproducibility by re-running our original analysis using the same LLMs. We assess replicability by repeating the analysis with three different, open-weight LLMs not used in the baseline study.

To test the *reproducibility* of our procedure using the same proprietary LLMs, we re-ran our entire analysis following a 3-month wait, using the latest versions of the same LLMs (see Supporting Information Appendix E, p. 22, for full details). The results, reported in the

left-hand panel of Table 6, show reproducibility far superior to any established human standard. In our reproduction run, correlations between pairs of LLM estimates ranged from .92 for the Environment to .96–.98 for all other issues except Decentralization, which replicated at “only” .69. This is not surprising given the noise in this issue, consistent with Table 3. These conform to standards for much computational research, when “bitwise reproducibility may be relaxed and reproducible results could be obtained within an accepted range of variation” (National Academies of Sciences, et al., 2019, p. 8). We conclude that our findings are reproducible.

To test *replicability* using a different data production method, we re-ran our entire analysis using entirely different LLMs. We picked three high-performing open-weight models: DeepSeek-V3, LLaMA-3.3, and Gemma-3. The results, reported in the right-hand panel of Table 6, are extremely encouraging. Comparing the two panels, the open weight models correlate with the expert benchmark only slightly less well than the proprietary models, as well as replicating the substantially lower correlations for decentralization. The central panel of Table 6 shows that correlations between the proprietary ensemble and the open-weight ensemble are very high, at about .95. Results generated by one class of LLM are replicating results generated by the other.

This is a particularly important finding for considerations of the costs and benefits of using open-weight or proprietary models. For this application at least, the benefits of using open-weight models—in terms of confidentiality, transparency, and replicability—seem to come at very little cost in terms of results. The key question concerns whether, notwithstanding inevitable stochastic components in the data generation process, the data generated from successive replications support the same general inferences. We find this is the case for our LLM-generated data.

## PREDICTIVE VALIDITY: LLM ESTIMATES OF COALITION ISSUE POSITIONS

Having substantively validated our LLM method using expert survey data, we use it to generate a dataset for which there is no expert benchmark. This measures policy positions of *coalition governments* on the six dimensions of interest, set out in their agreed policy declarations. Klüver et al. assembled a collection of 229 declarations issued by coalition cabinets in Eastern and Western Europe between 1945 and 2015 (Klüver et al., 2023). They manually unitized and labeled these using the MP coding scheme. We downloaded these agreements and scored them on our six dimensions of interest, using our  $3 \times 3$  design for both zero-shot learning as well as few-shot learning using the same prompt examples employed in the party manifesto analysis.

Limited overlap between start and end dates of each project, combined with our selection of party manifestos to coincide with years in which the benchmark expert surveys were fielded, left 23 coalition agreements in the Klüver et al. collection for which we also have LLM estimates of member party positions. Supporting Information Appendix Table F1 (p. 23) lists these. We used Klüver et al. labels to generate logit scales of coalition policy on the six dimensions of interest, using the method described in Supporting Information Appendix A2 (p. 4). For each coalition agreement, for each policy dimension of interest, we therefore have LLM and Klüver et al. estimates of *coalition* policy, and LLM and MP estimates of *member-party* policy positions.

We have no benchmark estimate of coalition policy positions. Indeed, there is no published estimate of these positions on the six issue dimensions of interest. We do, however, have a well-founded *theoretical* expectation about these. There are many different spatial models of government formation, dating back over 50 years of theoretical and empirical scholarship on coalition government. While these models differ in their precise predictions of agreed coalition policy, to the best of our knowledge, they all converge on one general prediction. On any given issue dimension, *coalition policy should be within the range of member parties' policy positions*. This is because, if coalition policy is *outside* the range of members' policy positions, then moving it into that range generates a Pareto improvement for all coalition members.<sup>23</sup>

We compare LLM-generated estimates with Klüver/MP hand-coded estimates of party and coalition policy positions against this theoretical expectation. Table 7 shows clear results. Except for decentralization, LLM-generated estimates of coal-

ition policy positions are significantly more likely to fall within the range of members parties' policy positions than those produced by MP-style hand-labeling.

It is not our purpose to assess the content validity of the hand-coding of coalition agreements using the MP scheme, and we cannot say that out-of-range estimates of coalition policy are definitively “wrong.” It is possible, for example, to imagine an idiosyncratic situation in which all member parties promise something completely infeasible in their manifestos—“deport all immigrants,” for example—that does not make it into a government declaration. But this caveat applies to both types of estimate. We find it encouraging, therefore, that LLM-generated estimates conform much more closely to theoretical expectations than those generated by methods based on labeling and counting sentences.

## CONCLUSION: THE REVOLUTIONARY POTENTIAL OF LLMs

Our aim is to evaluate a new method for analyzing political text, seeking reliable and valid inferences about the stated preferences of its author.<sup>24</sup> An ideal method would recruit a large sample of experts with profound knowledge of both the language and the political setting involved. These would be asked to read the relevant text for meaning, then respond to carefully crafted questions about the preferences of the author on issue areas of interest. We would need a large sample rather than a single expert because we expect experts to differ in their judgments about positions expressed in any given text.<sup>25</sup> This approach is infeasible for even a single text, let alone a large corpus of texts in many different languages dealing with many different political settings.

The current state of the art offers two approaches to reliable and replicable text analysis at scale. Quantitative TaDa methods are based on supervised machine learning algorithms of ever-increasing sophistication but are grounded in flawed assumptions about how quantitative patterns in text can be used to infer qualitative meaning. TaDa methods scale to massive text corpora and are now a tried and tested mainstream approach to analyzing political text—until recently, the only option for replicable text analysis at scale.

The new alternative is the portfolio of NLU tasks we assess here, implemented by LLMs. This extends the TaDa paradigm from statistical representation to interpretative modeling, generating results that are functionally equivalent to those produced by human

<sup>23</sup> Government formation models differ on *where within this range* they predict coalition policy.

<sup>24</sup> The systematic relationship between “stated” and fundamentally unobservable “true” preferences can only be established using theoretical models of politics which do not concern us here.

<sup>25</sup> Consider the dramatically varying expert interpretations of any classic political text.

**TABLE 7** Proportions of 23 coalition policy positions within the range of members' positions.

| Issue dimension  | LLM mean          |                  | Klüver and MP logit Scores |                  |
|------------------|-------------------|------------------|----------------------------|------------------|
|                  | Proportion inside | Missing <i>N</i> | Proportion inside          | Missing <i>N</i> |
| Economic         | .76               | 2                | .36                        | 1                |
| Social           | .75               | 3                | .45                        | 1                |
| Immigration      | .53               | 4                | .14                        | 9                |
| EU               | .57               | 0                | .45                        | 1                |
| Environment      | .78               | 0                | .18                        | 1                |
| Decentralization | .43               | 0                | .61                        | 5                |
| Overall          | .64               |                  | .38                        |                  |

Notes: Proportion inside gives the proportion of the relevant estimates for which the estimated coalition position is inside the range of coalition members' positions. "Missing *N*" denotes, for each dimension, the number of government declarations for which the LLMs or the Klüver/MP analysis returned a missing score. "LLM mean" is the proportion of coalition documents whose positions lie between the left-most and right-most coalition members, measured by LLMs; "Klüver and MP logit scores" are the same using logit scores from manifesto scoring counts, from the MP data and Klüver et al.

qualitative text analysis—but in many languages and at potentially massive scale. Casual use of LLMs to analyze political texts has increased rapidly with their exploding power and availability. Our concern here is to develop reliable and valid methods for using LLMs *as a scientific tool*. If shown to be reliable and valid, using LLMs to summarize and interpret texts by drawing on the collective wisdom ingested in their massive training data comes closest to the ideal of drawing on the collective wisdom of experts.

Our results demonstrate that LLMs can effectively perform this task. Correlations between LLM ensemble means and benchmark expert surveys are as high as we can reasonably expect for most issue dimensions, matching the upper bound of agreement between different expert surveys. Given a task for which there is no expert benchmark, the LLM ensemble produced coalition policy estimates with strong predictive validity, outperforming human-coded text labeling of the texts of coalition agreements.

Concerns about the scientific use of LLMs—to do with reproducibility, replicability, transparency, bias, and "leakage"—are understandable. We address the challenge of *reproducibility* by almost perfectly replicating all our results after a 3-month interval, using updated versions of each proprietary LLM. We evaluate *replicability* by rerunning our entire analysis with three completely different LLMs—open weight models with versions that can be downloaded and will not evolve over time. Again, we get almost identical results. These latter results are particularly significant since many scholars may have a principled preference for open weights rather than "closed" proprietary models, and we show here they work equally well.

Using "black box" LLMs to replicate qualitative text analysis is less *transparent* than quantitative TaDa methods, which implement explicit algorithmic or statistical procedures, typically using open-source software. Yet the cognitive processes of qualitative human coders and expert survey respondents are arguably even more opaque. While LLMs can be

prompted to provide detailed justifications for their outputs, human text annotators are almost never asked to do this, and would likely find it challenging if asked to do so.

A related and distinctive contribution of our approach is the production of concise policy summaries in the first stage of our analysis. In contrast to sentence-level classification approaches, which segment text into isolated labeling decisions, our method begins with a holistic summarization stage that yields interpretable, document-level representations of policy content. These summaries are themselves a valuable research output: They not only enhance interpretability and transparency but also provide a reusable textual representation of policy content that could support further analysis. Our results—and the extensive replication materials accompanying this paper—should encourage future work on using, evaluating, and standardizing such summaries.

While LLMs may still operate as black boxes, we have shown that the summary-and-scaling pipeline yields outputs that are systematic, replicable, and substantively valid. An ensemble of models, applied across diverse issue areas, consistently maps textual inputs to policy position estimates in a reliable and interpretable manner.

To evaluate potential *bias*, we need a ground truth against which bias can be measured—something fundamentally unobservable when it comes to the "true" issue positions of politicians. Using ensembles of experts as a surrogate for ground truth, we found strong evidence of "bias" for decentralization, some evidence of this for environmental policy, and none for the other four issue areas. LLMs tended to conclude that party manifestos articulated systematically more pro-decentralization and pro-environmental policies than those identified by experts. However, when comparing LLM estimates to those generated by the MP, this seems to reflect "structural bias" inherent in manifesto content rather than bias attributable to LLMs.

Substantially lower correlations between LLM estimates and the expert benchmark for decentralization policy supplement other strong evidence against the possibility of “leakage,” in which LLMs retrieve information from their training data rather than analyzing the manifestos de novo. If that were not the case, there would be no reason for the models to perform markedly worse for decentralization than for the five other issue areas we study.

Developing scientifically valid methods for using LLMs in political text analysis is crucial because these powerful tools are widely accessible, easy to use, multilingual, and cheap. Their potential is transformative, enabling motivated scholars, regardless of their geographic locations or funding constraints, to conduct sophisticated, targeted analyses without the massive resources traditionally required for large-scale qualitative text analysis. We have demonstrated how, with minimal resources and in a short timeframe, it is possible to replicate the outputs of extremely expensive and time-consuming projects reliant on human experts, such as the MP and CAP. By leveraging LLMs for automated text analysis, underfunded junior researchers can generate the specific text data they need for their work, rather than relying on legacy datasets of limited relevance.

We offer a demonstration rather than a technical analysis of LLMs and their potential in the analysis of political text. Using state-of-the-art LLMs available at the time of writing, we achieved results comparable to those from projects reliant on human experts. By the time this article is published, ongoing advances in this rapidly moving field will likely have produced a new generation of LLMs that are not only significantly more powerful than these but also even easier to use. The strong results we report here will only get stronger as LLMs become more powerful. Love them or hate them, LLMs are poised to drive a new measurement paradigm in political science, making it essential to establish clear and robust protocols to ensure their valid and replicable use.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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